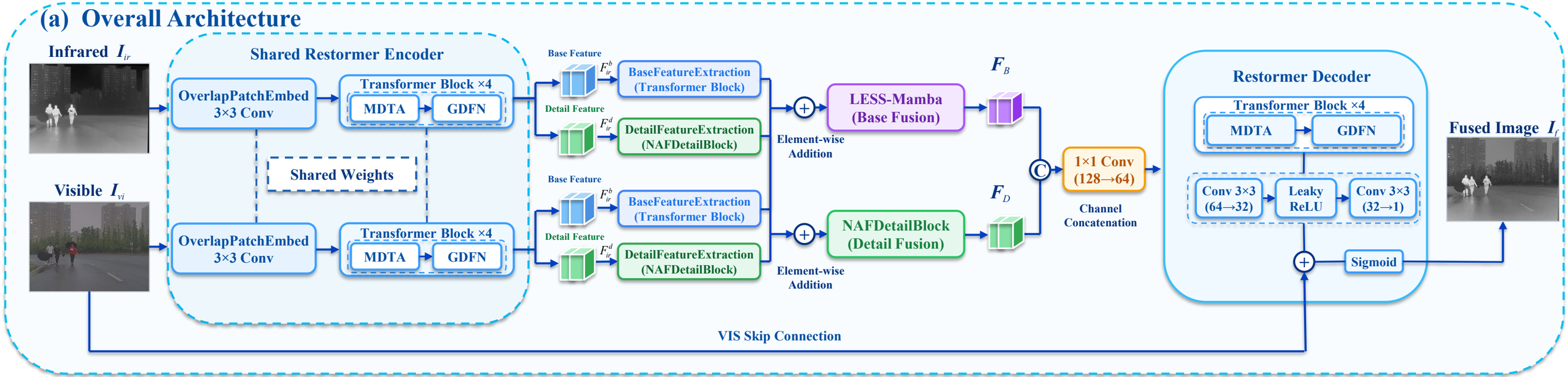

(a) Overall Architecture
Infrared I_{ir}
Visible I_{vi}
Shared Restormer Encoder
OverlapPatchEmbed 3×3 Conv
Transformer Block ×4
MDTA
GDFN
Shared Weights
OverlapPatchEmbed 3×3 Conv
Transformer Block ×4
MDTA
GDFN
Base Feature
Detail Feature
F_{ir}^{b}
F_{ir}^{d}
BaseFeatureExtraction (Transformer Block)
DetailFeatureExtraction (NAFDetailBlock)
Base Feature
Detail Feature
F_{ir}^{b}
F_{ir}^{d}
BaseFeatureExtraction (Transformer Block)
DetailFeatureExtraction (NAFDetailBlock)
Element-wise Addition
LESS-Mamba (Base Fusion)
NAFDetailBlock (Detail Fusion)
Element-wise Addition
F_B
F_D
C
1×1 Conv (128→64)
Channel Concatenation
Restormer Decoder
Transformer Block ×4
MDTA
GDFN
Conv 3×3 (64→32)
Leaky ReLU
Conv 3×3 (32→1)
Sigmoid
Fused Image I_f
VIS Skip Connection